GLS University
Faculty Of Computer Applications & Information Technology
iMScIT/ Semester IX

Capstne Project - 2026-27

Student 1 Name: SAIYED MOHAMMED HAMZAH KHALIDMIYA	**Mobile : 9737075395**	**Enroll : 202201619010072**	**Div_Rollno. A-74**
Student 2 Name: RAVAL VED SANJAYBHAI	**Mobile : 8320381461**	**Enroll : 202201619010071**	**Div_Rollno. A-73**

Project Title: AI-Powered Intelligent Crop Health & Precision Farming Platform

Frontend Technologies : React.js, HTML5, CSS3, JavaScript, Tailwind CSS

Backend Technologies: Python, FastAPI, PostgreSQL, TensorFlow, OpenCV, YOLOv8

List of Hardware / Sensor / Processor (if Project on Embedded System) :

Project Description: AI-Powered Intelligent Crop Health & Precision Farming Platform is a web-based Artificial Intelligence application designed to support precision agriculture by helping farmers monitor crop health and make data-driven decisions. The platform uses Computer Vision and Machine Learning to detect plant diseases and pests from leaf images, analyze soil health, predict crop yield, recommend smart irrigation schedules, and provide personalized farming guidance through an AI-powered chatbot. By integrating real-time analytics, predictive models, and intelligent recommendations into a single platform, the system aims to improve crop productivity, optimize resource utilization, and promote sustainable farming practices.

Project Modules:

Module 1: Data Acquisition & Preprocessing
Functionalities:

- Upload plant leaf images.
- Enter crop and farm details.
- Input soil parameters (pH, NPK, Moisture).
- Fetch weather data using API.
- Image preprocessing (resize, normalization).
- Feature extraction using YOLOv8.

Description:
This module collects and preprocesses crop, soil, and weather data. It prepares the input data required for AI models to perform accurate crop health analysis and prediction.

Module 2: Plant Disease & Pest Detection
Functionalities:

- Detect plant diseases from leaf images.
- Identify pests using Computer Vision.
- Display confidence score.

- Highlight affected regions.
- Store prediction history.

Description:
This module uses AI-based Computer Vision models to identify plant diseases and pests from uploaded crop images. It provides prediction results with confidence scores to assist farmers in early disease detection.

Module 3: Soil Analysis & Smart Farming Recommendations
Functionalities:
- Analyze soil health.
- Recommend fertilizers.
- Suggest suitable crops.
- Recommend irrigation schedules.
- Display weather-based farming alerts.

Description:
This module analyzes soil and environmental conditions to generate intelligent farming recommendations. It helps farmers optimize fertilizer usage, irrigation planning, and crop selection.

Module 4: Crop Yield Prediction & Farm Analytics
Functionalities:
- Predict crop yield.
- Analyze historical prediction data.
- Generate farm performance analytics.
- Create graphical reports.
- Export reports in PDF format.

Description:
This module predicts expected crop yield using Machine Learning models and presents farm statistics through interactive dashboards and reports for better decision-making.

Module 5: AI Chatbot & System Integration
Functionalities:
- AI Agronomist Chatbot.
- Farmer profile management.
- User authentication.
- Admin dashboard.
- FastAPI backend integration.
- PostgreSQL database management.
- Prediction history management.

Description:
This module integrates all system components into a unified platform. It provides AI-powered farming assistance through a chatbot while managing users, prediction history, reports, and secure backend operations.